

Laboratory Experiment No. 1

Objective: To construct and verify the working of half adder, full adder, and 4-bit adder circuits employing fundamental logic gates, and to realize half adder and full adder designs using exclusively NAND gates.

1. Half Adder Circuit (Basic Logic Gates)

A half adder is a fundamental combinational logic circuit that performs binary addition of two single-bit inputs. It generates two outputs — the Sum (S) and the Carry (C).

Boolean Expressions:

$$S = A \oplus B \quad (\text{Implemented using XOR gate})$$

$$C = A \cdot B \quad (\text{Implemented using AND gate})$$

The XOR gate yields a logic HIGH when both inputs carry different values, whereas the AND gate activates only when both inputs are HIGH, generating the carry output.

2. Half Adder Circuit (NAND Gate Realization)

Since the NAND gate is a universal gate, it is capable of implementing any Boolean function without the need for other gate types. The half adder can therefore be entirely redesigned using only NAND gates.

Key points of NAND-based implementation:

- The XOR function required for the Sum output is constructed using a combination of NAND gates.
- The AND function required for the Carry output is equivalently obtained through NAND gate logic.
- No additional gate types are needed — the entire circuit uses only NAND gates.

This demonstrates the universal nature of the NAND gate, enabling complete circuit design without requiring dedicated XOR or AND components.

3. Full Adder Circuit (Basic Logic Gates)

A full adder extends the functionality of a half adder by incorporating a third input — the carry-in (Cin). It produces a Sum (S) and a Carry-out (Cout) as outputs.

Output Equations:

$$S = A \oplus B \oplus C_{in}$$

$$C_{out} = (A \cdot B) + (C_{in} \cdot (A \oplus B))$$

Structurally, a full adder is built by cascading two half adders followed by a single OR gate to compute the final carry-out.

4. 4-Bit Binary Adder (Basic Gates)

A 4-bit adder is designed to add two 4-bit binary numbers simultaneously. It is formed by chaining four full adders in sequence, where the carry output of one stage feeds directly into the carry input of the next.

Operational characteristics:

- Every full adder stage processes its respective pair of input bits together with the carry signal from the preceding stage.
- The initial carry-in (Cin) for the least significant full adder is set to logic 0.
- The carry ripples sequentially from one stage to the next — this topology is referred to as a Ripple Carry Adder (RCA).

The circuit delivers a 4-bit sum output (S1–S4) along with a final carry-out (Cout).

Logisim Implementation — Circuit Diagrams

The following diagrams illustrate the Logisim simulation of the designed circuits:

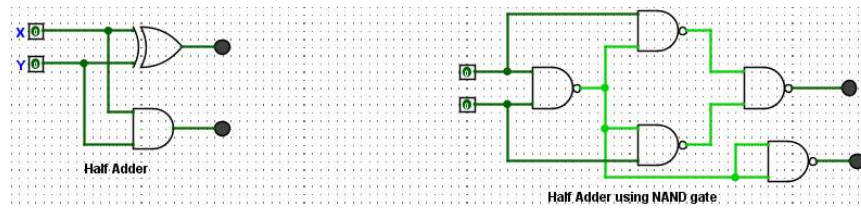


Figure 1: Half Adder and Full Adder — Basic Gates & NAND Gate Implementations

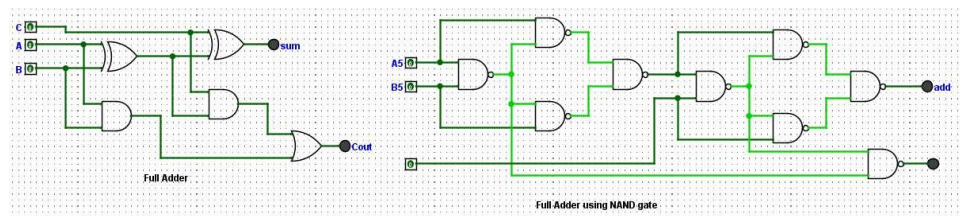


Figure 2: 4-Bit Ripple Carry Adder in Logisim

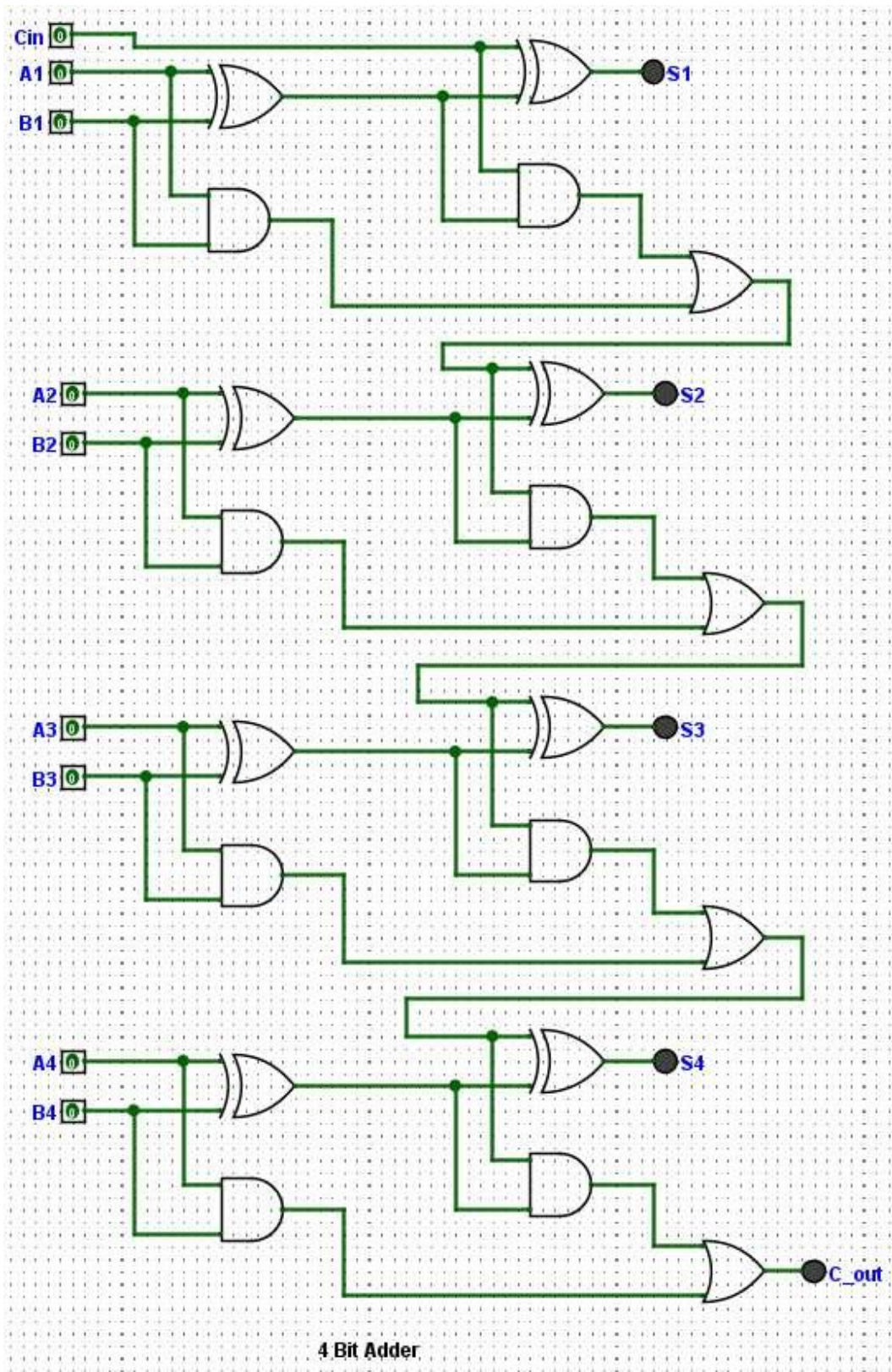


Figure 3: Complete 4-Bit Adder Circuit